Lab 5: Automation — Who or What Decides?

ECON 470 — Microeconomics of Artificial Intelligence

Prof. Eduardo Zambrano, Cal Poly

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Groups: 2–3 students. **Only one submission per group, please.**

Due: Wednesday, June 3, 11:59 PM.

Submit: A writeup with your answers to the numbered questions below. Show your work for all calculations.

Points: 100 total. See point values next to each problem.

Time estimate: 3–4 hours. The handout does most of the model-building; the lab pushes you to apply, perturb, and interpret.

What you need: The handout *Who Decides? From One Decision to a Spectrum of Tasks* (in hand from class), Gans Chapter 6 §6.3, and a calculator. No coding required.

If you get stuck: The Prediction Tutor can help you work through problems without giving away the answer: https://chatgpt.com/g/g-6a19dad4c110819194998f54e727156f-automation-tutor. Use it sparingly — you'll learn more by wrestling with the questions yourself first.

Notation throughout this lab. All variables and assumptions are as defined in the handout: N states $\theta \in \{1, \ldots, N\}$ with prior $p > 1/N$ on the default; per-decision match payoff R and mismatch payoff 0; AI accuracy $e \in (p, 1]$; coding cost C (smart machine); delay discount $\delta < 1$ and error-mitigation share $\alpha \in [0, 1]$ (human-with-AI). The three regime payoffs are

$$\text{Rule: } pR, \qquad \text{Smart machine: } eR - C, \qquad \text{Human: } \delta\big[e + (1-e)\alpha\big]R.$$

Running parameters (Part I). $R = 100$, $p = 0.6$, $\delta = 0.75$, $\alpha = 0.7$, $C = 10$ (so $C/R = 0.10$).

Running parameters (Part II). Same constants, with $e(j) = 1 - kj$ for $j \in [0, 1]$.

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1 Three Regimes, One Decision [30 pts]

This problem works with Part I of the handout: a single decision at a single AI accuracy e.

(a) At the running parameters and $e = 0.65$, compute the per-decision payoff under each of the three regimes. *(Report all three numbers.)*

(b) Which of the three regimes is optimal at $e = 0.65$? Which regimes beat the rule? Which do not? *(Use your numbers from (a) to answer.)*

(c) The handout derives three pairwise boundaries:

$$\text{B1 (smart vs. rule): } \frac{C}{R} \leq e - p,$$

$$\text{B2 (human vs. rule): } \delta[e + (1-e)\alpha] \geq p,$$

$$\text{B3 (smart vs. human): } \frac{C}{R} \leq e(1-\delta) - \alpha\delta(1-e).$$

For each boundary, write down the corresponding threshold value of e (call them e_1, e_2, e_3) holding the other parameters at their running values. That is, find the smallest e for which each boundary is satisfied with equality.

(d) Order e_1, e_2, e_3 from smallest to largest. Use that ordering to describe, in words, what happens as e rises from 0 to 1: which regime wins in which range of e?

(e) The handout (Boundary 3) shows that the threshold for the smart machine to beat the human is *increasing* in e. State in your own words what this means for the human decision-maker as AI accuracy improves on a *fixed* task. *(One or two sentences.)*

(f) Now consider a second task whose mitigation parameter is $\alpha = 0.3$ rather than 0.7, with all other parameters unchanged. At $e = 0.65$, which regime is optimal? Why does the answer differ from (b)? *(One sentence.)*

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2 Who Decides in $(\alpha, C/R)$ Space [30 pts]

This problem works with the handout's "who decides" diagram at fixed $e = 0.65$ (figure following Boundary 3 in the handout).

(a) At the running parameters and $e = 0.65$, the three boundaries meet at the point $(\alpha^*, C/R^*)$. Find the value $(\alpha^*, C/R^*)$.

(b) Consider a task with $\alpha = 0.6$ and $C/R = 0.03$. Use the three boundaries to determine which regime is optimal. *(You may either plug into the boundaries or read off the diagram — but show the inequality check that pins down your answer.)*

(c) Now hold $\alpha = 0.6$ fixed and ask: at what value of C/R does the optimal regime switch from smart machine to human? *(Use B3, since that is the binding boundary between those two regimes at this α.)*

(d) The handout notes that *as e rises*, the smart-vs-human diagonal (B3) shifts upward and flattens. Recompute the diagonal's vertical intercept (its value at $\alpha = 0$) at $e = 0.65$ and at $e = 0.85$. By how much does the intercept rise?

(e) Pick a task at $(\alpha, C/R) = (0.6, 0.10)$. At $e = 0.65$, which regime is optimal? At $e = 0.85$, which regime is optimal? Use your boundary calculations to justify the switch (or non-switch). *(This is the "displacement at the margin" claim made concrete.)*

(f) The handout argues that the §6.3 model does **not** license a global claim of the form "as e rises, the human region shrinks." Explain in 2–3 sentences why not, in terms of what the model does and does not pin down.

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3 The Linear Specification and the Biphasic Pattern [40 pts]

This problem works with Part II of the handout: $e(j) = 1 - kj$ for $j \in [0, 1]$, with the same constants as Part I.

The three cutoffs in j derived in the handout are:

$$j_{SR}(k) = \frac{1 - p - C/R}{k}, \quad j_{HR}(k) = \frac{\delta - p}{\delta\, k\, (1 - \alpha)}, \quad j_{SH}(k) = \frac{1 - \delta - C/R}{k\,[1 - \delta(1 - \alpha)]}.$$

(a) Verify that at the running parameters, the three cutoff *coefficients* (the numerators divided by the constants in the denominators) are 0.30, 0.667, and 0.194 respectively. *(Show each substitution. These are the "$1/k$-scaled" coefficients reported in the handout.)*

(b) At $k = 1.2$, compute $j_{SH}(1.2)$ and $j_{HR}(1.2)$. Write out the assignment ladder: which complexity range goes to the smart machine, which to the human, which to the rule? What is the width of the human region at this k?

(c) Now let the AI improve: k falls from 1.2 to 0.8. Recompute $j_{SH}(0.8)$ and $j_{HR}(0.8)$. Is $j_{HR}(0.8) \leq 1$ or > 1? What is the width of the human region now?

(d) Continue: let k fall further to 0.5. Recompute both cutoffs. Now check: does $j_{HR}(0.5) \leq 1$? If $j_{HR}(0.5) > 1$, *cap* the human region at $j = 1$, so the human region is $[j_{SH}(0.5),\, 1]$. Report the (capped) width.

(e) From parts (b)–(d), what does the width of the human region do as k falls from $1.2 \to 0.8 \to 0.5$? Is this consistent with the *biphasic pattern* described in the handout's Proposition? Identify which phase (expansion or contraction) each step is in.

(f) Solve for the critical value k^* at which $j_{HR}(k^*) = 1$ — the boundary between the expansion phase and the contraction phase. *(Set $j_{HR}(k) = 1$ and solve for k.)* Then explain in 1–2 sentences why k^* is exactly the peak of the human region's width: what changes at $k = k^*$?

(g) **Interpretation.** The handout connects this biphasic pattern to the "radiologist puzzle" (Hinton 2016: stop training radiologists; 2026: more radiologists than ever). In 3–4 sentences, explain which phase the radiology profession appears to be in, and what the model predicts will happen if AI accuracy continues to improve. Be specific about which cutoff would have to bind for the contraction phase to begin, and what that means for the *complexity* of the tasks radiologists handle.

(h) **A different industry.** Pick one of: image labeling, legal contract review, software debugging, transcription. In 2–3 sentences, argue whether the *task-complexity ceiling* (the maximum j in the industry) is high or low, and use that to predict which phase the industry is more likely to be in as AI improves. (You will not be graded on which industry you pick or which phase you predict — you will be graded on whether your argument cleanly uses the Part II machinery.)